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Mechanism of limitation of pronation/supination of the forearm in geometric models of deformities of the forearm bones

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Abstract

Objective. To clarify the mechanism of limitation of pronation/supination of the forearm associated with the angular deformity of the forearm bones and narrowing of the interosseous space.

Design. A three-dimensional geometric model of the forearm bones with the interosseous membrane and its axial section were used.

Background. Limitation of pronation/supination associated with the deformity of the forearm bones is one of the significant problems encountered in the treatment of the forearm fracture. Elucidation of its mechanism is important for its prevention and treatment.

Methods. In the axial section, the effects of the positional relationship between the axis of pronation/supination and the forearm bones on the range of pronation/supination was studied using analytic geometry in each model with non-narrowing or narrowing of the interosseous space. Subsequently, in the three-dimensional model each forearm bone with the same angular deformity, the direction and magnitude of the angular deformity which would lead to limitation of pronation/supination were calculated using analytic geometry. Each parameter of the models was obtained by the radiographic measurements of the normal forearms.

Results. When the axis of the pronation/supination passed through the interosseous region (less than 2 cm radioulnarly and 0.8 cm anteroposteriorly) in the model of the axial section without narrowing of the interosseous space, more than 40° of pronation and supination were possible. When the axis deviated from this region, significant loss of pronation/supination was observed associated with restriction by the interosseous membrane rather than impingement. Furthermore, the area of this region decreased according to narrowing of the interosseous space with shortening of the interosseous membrane. In the three-dimensional model, the direction and magnitude of the angular deformity which would lead to significant loss of pronation/supination was more than 14° radially, 7° ulnarly, 5° anteriorly, 4° posteriorly.

Conclusions. The positional relationship between the axis of pronation/supination and the forearm bones with the interosseous membrane may play an important role regarding pronation/supination of the forearm.

Relevance

Evaluation of the bone deformities based on understanding this mechanism of limitation of pronation/supination would lead to an appropriate treatment of malunion of the forearm bones.
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Keywords: Biomechanics; Forearm; Geometric model; Interosseous membrane; Malunion; Rotation

1. Introduction

Limitation of pronation/supination of the forearm (forearm rotation) is one of the problems often encountered in the treatment of fracture of the forearm bones. Angular deformity of the forearm bones and narrowing of the interosseous space between the radius and ulna lead to limitation of forearm rotation [1–4]. In previous studies about the effects of angular deformity of the forearm bones on forearm rotation [1,4], increasing tension of the interosseous membrane (IOM) and impingement between the forearm bones were

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reported as the disturbances which cause limitation of forearm rotation. However, the direction and magnitude of the angular deformity which would lead to such disturbances have not been determined. We have previously described that the locus of the top of the convexity of the radius during forearm rotation in the cases with angular deformity of the radius would be determined by the position of the axis of forearm rotation (AFR) [5]. In the cases in which the interosseous space widens during forearm rotation, the rotation may be limited by the IOM when the interosseous space reaches the length of the IOM [2,5]. We thought the positional relationship between the AFR and the forearm bones would affect the range of forearm rotation. And we considered that one positional relationship between them might correspond to one combination of the direction and magnitude of the angular deformity, when both forearm bones were deformed in the same direction and to the same magnitude at a certain position. On the basis of this idea, we studied the direction and magnitude of the angular deformity which would lead to limitation of forearm rotation. First, we estimated the region of the AFR which would allow the radius to pronate and supinate without being restricted by the IOM or impingement on the ulna in a two-dimensional (2D) model of an axial section of the forearm. Second, the effects of narrowing of the interosseous space between the radius and ulna on forearm rotation were studied in this 2D model. Third, based on the results of the 2D geometric study, the direction and magnitude of the angular deformity which would lead to limitation of forearm rotation were studied in a three-dimensional (3D) forearm model.

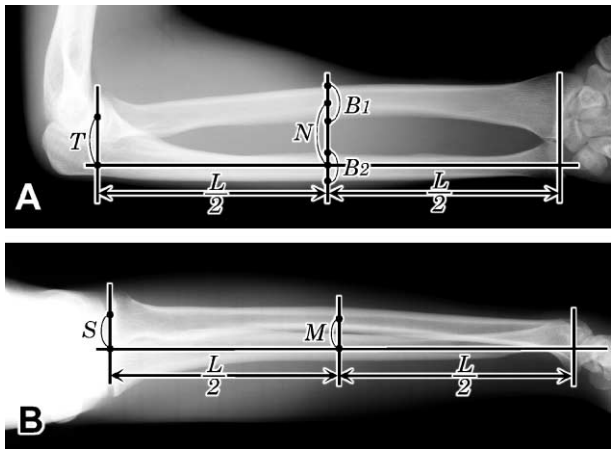


Fig. 1. Measurements of posteroanterior and lateral radiograms of the left forearm. (A) Posteroanterior view which was printed in reverse in order to match the viewpoint with the models of this study; (B) lateral view. L, distance from the distal end of the ulna to the perpendicular line from the proximal end of the radial head to the longitudinal axis of the ulna; S and T, distance between the longitudinal axis of the ulna and the center of the proximal end of the radius; M and N, distance between the longitudinal axis of the ulna and the center of the shaft of the radius at the middle of the forearm; B_1 and B_2, width of the shaft of the radius and ulna respectively.

2. Methods

2.1. Radiographic measurements of the forearm bones

Radiographic measurements were performed on biplane radiographs of the left forearms of five male volunteers without bony abnormalities to design a geometric model of the forearm bones. Their mean age was 40.0 (SD, 4.6) years, mean height 173.2 (SD, 5.6) cm and mean body mass 69.2 (SD, 11.6) kg. Posteroanterior and lateral radiographs of the forearm were obtained by positioning the shoulder in abduction of 90°, the elbow in flexion of 90°, and the forearm in neutral forearm rotation. The forearm was placed on the film cassette and the tube-film distance was set at 1 m. The longitudinal axis of the ulna on the radiogram was set at the middle third of the ulna. In the posteroanterior view, we measured the length of the longitudinal axis of the ulna from the distal end of the ulna to the perpendicular line from the proximal end of the radius (L), the distance between the longitudinal axis of the ulna and the center of the proximal end of the radius (T), the distance between the longitudinal axis of the ulna and the center of the shaft of the radius at the middle of the forearm (N), and the width of the shafts of the radius and ulna at the middle of the forearm (B_1 and B_2). Half of the average of B_1 and B_2 was represented by B. In the lateral view, we measured the distance between the longitudinal axis of the ulna and the center of the proximal end of the radius (S) and the distance between the longitudinal axis of the ulna and the center of the shaft of the radius at the middle of the forearm (M) (Fig. 1).

2.2. Three-dimensional forearm model

The model for this study was the left upper extremity. The longitudinal axis of the humerus was fixed in the vertical position and the elbow was held at 90° of flexion. The initial rotation of the forearm was adjusted in the neutral position. The forearm bones were assumed to be straight. The attachments of the IOM to the ulna and those to the radius were assumed to be collinear and these lines were defined as line U and line R. The plane through the proximal end of the radius perpendicular to line U was defined as plane P. The distance between plane P and the distal end of the ulna was represented by L. The horizontal and vertical components of the distance between the longitudinal axes of the forearm bones at plane P were represented by S and T respectively, and those at the point $L/2$ distal from plane P by M and N respectively. The AFR was defined as the line which passes through the center of the proximal end of the radius and the distal end of the ulna as described

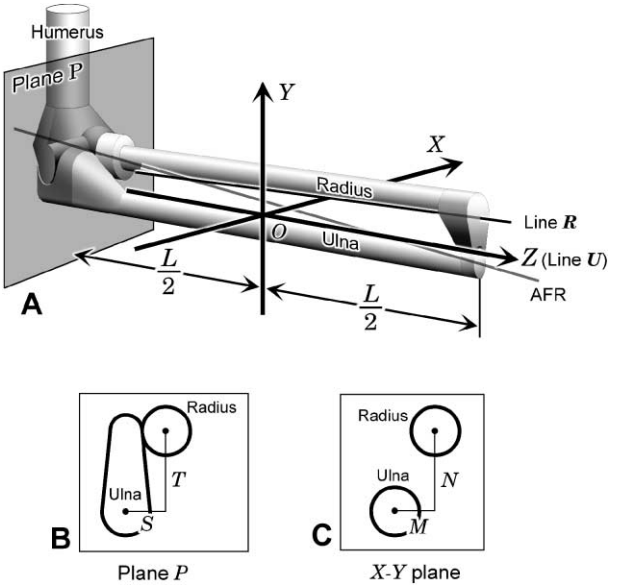


Fig. 2. (A) Schematic representation of the model of left forearm bones with orthogonal 3D-coordinates (X, Y, Z). (B,C) Schematic representation of the sections through plane P and XY-plane; L, length between plane P and the distal end of the ulna; AFR; axis of the forearm rotation; S and T, horizontal and vertical components of the distance between the longitudinal axes of the forearm bones at plane P; M and N, the same components at XY-plane.

elsewhere [6]. When the forearm bones are deformed without dislocation of the adjacent joints, a line defined in the same manner as that used in normal cases might serve to represent the AFR. A 3D-right-handed orthogonal coordinate system (X, Y, Z) with its origin at point $L/2$ distant from plane P on line U was established in this model. The Y-axis was parallel to the longitudinal axis of the humerus, directed upward; the Z-axis coincided with line U, directed distally; the X-axis was perpendicular to both the Y- and Z-axes (Fig. 2).

2.3. Two-dimensional model of the axial section of the forearm

We established a 2D-orthogonal coordinate system (x,y) in the sectional plane perpendicular to the AFR, with its origin coinciding with that of the 3D-coordinate system (Fig. 3). The y-axis extended from the origin to the attachment of the IOM on the radius and the x-axis was positive in the posterior direction of the forearm. The sections of the radius and the ulna were modified to circles with a diameter of $2 \times B$ in order to simplify the equations which described later, and these two sections were represented by circle R and circle U, respectively. The intersections where the xy-plane met the AFR and line R were defined as points C and A respectively, and the coordinates of these points in the xy-plane were assumed to be $C\ (C_x, C_y)$ and $A\ (0, a)$, the value of a can be expressed using the values of L, M, N, S, T and B.

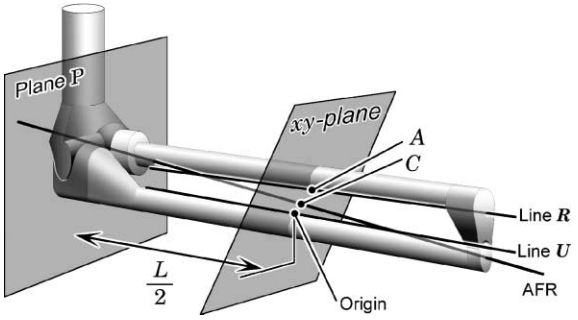


Fig. 3. Schematic representation of the model of the forearm bones showing the AFR and the sectional plane (xy-plane). A, intersection where the sectional plane meets the line R; C, intersection where the sectional plane meets the AFR.

The center of each circle, the origin, and point A were adjusted to be collinear. The counterclockwise rotation of circle R around point C represented pronation of the forearm and the clockwise rotation supination (Fig. 4). The angle of a counterclockwise rotation of circle R was assigned a positive number, and the angle of a clockwise rotation a negative number. The IOM was represented as a string attached to the origin and point A. The length of the string was represented by d, which was assumed to allow for slackness in addition to the length between the origin and point A. The slackness of the string was set at 6% of the length between the origin and point A on the basis of a study by Pfaeffle et al. [7]. They described the stress-strain curve of the central band which is the most dominant structure of the IOM, and reported that the strain at failure was 9.0%. We selected the value near the

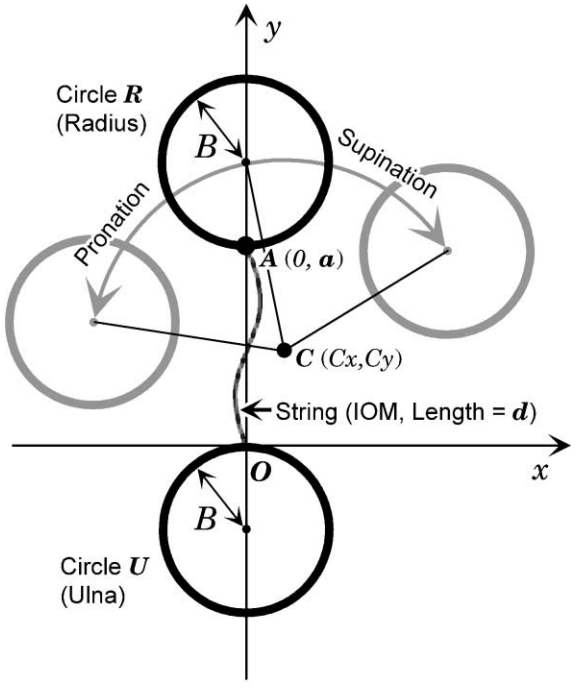


Fig. 4. Orthogonal 2D-coordinate system (x,y) in the sectional plane.

upper limit in the linear portion of the stress-strain curve as the slackness of the string for this study.

2.4. Geometrical study

2.4.1. Study 1 (2D geometric study)

The region of point C around which circle R was allowed to rotate without being restricted by the string or impingement on circle U was studied in the xy-plane. When circle R was allowed to rotate from neutral to more than ρ° of supination or more than σ° of pronation, the region of point C was defined as function $\mathrm{Reg}(-\rho)$ or $\mathrm{Reg}(\sigma)$, respectively. The intersectional region between $\mathrm{Reg}(-\rho)$ and $\mathrm{Reg}(\sigma)$ was defined as $\mathrm{Reg}(-\rho/\sigma)$. $\mathrm{Reg}(\phi)$ for $0^\circ \leqslant \phi \leqslant 90^\circ$ was enclosed by the graphs of the equations which express the following four critical cases: (1) the distance between the points of attachment of the string is attained the maximum halfway the rotation of circle R, and the maximum distance equals the length of the string; (2) the distance between the points of attachment of the string is attained the maximum at the end of the rotation of circle R, and the maximum distance equals the length of the string; (3) the distance between circle R and circle U is attained the minimum halfway the rotation of circle R, when the circles contact each other; and (4) the distance between circle R and circle U is attained the minimum at the end of the rotation of circle R, when the circles contact each other. The boundary of the region of point C in case (1) was expressed by an ellipse

$$\{2aC_y + d^2 - a^2\}^2 = 4d^2(C_x^2 + C_y^2). \tag{1}$$

The boundary of the region of point C in case (2) was expressed by a graph of the parametric equations with parameter θ. These equations were expressed using matrices,

$$\begin{pmatrix} C_x \\ C_y \end{pmatrix} = \frac{1}{2}\csc\frac{\Phi}{2}\left\{ \begin{pmatrix} \cos\frac{\Phi}{2} & 0 \\ 0 & \sin\frac{\Phi}{2} \end{pmatrix} \begin{pmatrix} a + BV_1 \\ a + BV_2 \end{pmatrix} + \begin{pmatrix} \sin\left(\Theta - \frac{\Phi}{2}\right) & \cos\left(\Theta - \frac{\Phi}{2}\right) \\ -\cos\left(\Theta - \frac{\Phi}{2}\right) & \sin\left(\Theta - \frac{\Phi}{2}\right) \end{pmatrix} \begin{pmatrix} BV_3 \\ BV_4 - d \end{pmatrix} \right\} \tag{2}$$

where

$$\begin{cases} \Phi = \phi\pi/180 \\ \Theta = \theta\pi/180 \end{cases}$$

and

$$\begin{cases} V_1 = 2, & V_2 = 0, & V_3 = 2, & V_4 = (\phi - 2\theta - 180)\pi/180 & (\theta \leqslant -90) \\ V_1 = 1, & V_2 = 1, & V_3 = 1, & V_4 = (\phi - \theta - 90)\pi/180 & (-90 \leqslant \theta \leqslant \phi - 90) \\ V_1 = 0, & V_2 = 0, & V_3 = 0, & V_4 = 0 & (\phi - 90 \leqslant \theta \leqslant 90) \\ V_1 = 1, & V_2 = -1, & V_3 = -1, & V_4 = (\theta - 90)\pi/180 & (90 \leqslant \theta \leqslant \phi + 90) \\ V_1 = 2, & V_2 = 0, & V_3 = -2, & V_4 = (-\phi + 2\theta - 180)\pi/180 & (\phi + 90 \leqslant \theta) \end{cases}$$

The boundary of the region of point C in case (3) was expressed by a hyperbola

$$\{2(a + 2B)C_y - (a + B)^2 + 5B^2\}^2 = 16B^2\{C_x^2 + (C_y + B)^2\}. \tag{3}$$

The boundary of the region of point C in case (4) was expressed by a circle

$$\left(C_x - \frac{(a + 2B)\sin\Phi}{2(1 - \cos\Phi)}\right)^2 + \left(C_y - \frac{a}{2}\right)^2 = \frac{2B^2}{(1 - \cos\Phi)} \text{ where } \Phi = \phi\pi/180. \tag{4}$$

$\mathrm{Reg}(-\phi)$ for $0^\circ \leqslant \phi \leqslant 90^\circ$ was a reflection of $\mathrm{Reg}(\phi)$ on the y-axis. The boundary lines of $\mathrm{Reg}(\phi)$ and $\mathrm{Reg}(-\phi/\phi)$ at varying values of ϕ, with ϕ changing from 10° to 90° at 10° intervals, were drawn.

2.4.2. Study 2 (2D geometric study)

The effects of a reduction ofthe distance between circle R and circle U on the size of $\mathrm{Reg}(-70^\circ/70^\circ)$ were studied using the 2D model and the equations used for study 1. The distance between circle R and circle U was represented by a and the length of the string by d (Fig. 4). The initial values of a and d were obtained using the results of radiographic measurements. We divided this study into two, study 2A and study 2B according to the behavior of the string while gradually reducing a. In study 2A, the length of the string was fixed at the initial value of d. In study 2B, the length of the string was shortened in proportion to the reduction of a and its length was d which was 6% longer than each a. This shortening of the string simulated the scar formed along the IOM or a similar occurrence. The boundary lines of $\mathrm{Reg}(-70^\circ/70^\circ)$ at varying values of a, while the subtrahends from the initial value of a which were increased from 0 to 10 mm by 2.5 mm each time, were drawn. The size of $\mathrm{Reg}(-70^\circ/70^\circ)$ at the initial value of a was defined as the initial size of that.

2.4.3. Study 3 (3D geometric study)

The direction and magnitude of angular deformities of the forearm bones that allowed the radius to pronate and supinate without being restricted by the IOM or impingement on the ulna were studied in a 3D forearm

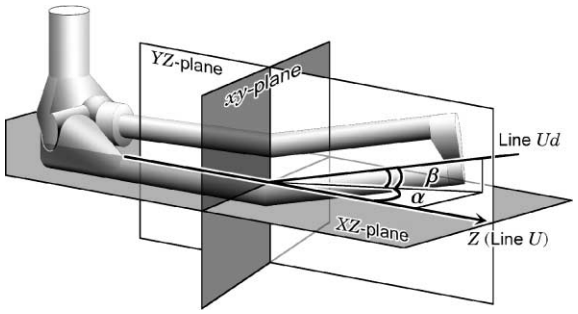


Fig. 5. Schematic representation of the model of forearm bones with angular deformity.

model. The same direction and magnitude of angular deformity were established for each forearm bone at the intersection points where the bones met on the *xy*-plane. The points of attachment of the IOM on the distal half of the ulna were defined as the line *Ud*. The angle between the *YZ*-plane and line *Ud*, and that between the *XZ*-plane and the line *Ud* were represented by α and β respectively (Fig. 5). When the origin of the *xy*-plane was moved along line *U* from one end to the other end of the forearm, the space would be constricted by connection of each Reg($-\phi/\phi$) for a certain ϕ. This space was defined as region *D*. Region *D* of the forearm model without bone deformity was a mathematically convex space. Provided the shape of the forearm bones allows the radius to rotate freely, the AFR would exist in region *D*. When the forearm bones angulate in the *xy*-plane, region *D* is composed of the two convex spaces which contact at the *xy*-plane. Deviation of the AFR from region *D*, if there were any, would occur at the *xy*-plane. Therefore, examination of the relationship between the position of the AFR and Reg($-\phi/\phi$) in the *xy*-plane would indicate whether the AFR exists within region *D*. Angles α and β could then be obtained by simultaneously solving the equations which express Reg($-\phi/\phi$) and the AFR. The range of the permissible angle of angular deformity was represented by function Ang($-\phi/\phi$) when the radius was allowed to rotate ϕ or more of supination and pronation. The boundary lines of Ang($-\phi/\phi$) at varying values of ϕ, while increasing ϕ from 10° to 90° at 10° intervals, were drawn. The average values of *L*, *M*, *N*, *S*, *T* and *B*, which were obtained by radiographic measurements were substituted the equations. Mathematica version 3.0 (Wolfram Research, Champaign, IL, USA) was used to simultaneously solve the equations in study 3 and to draw the graphs resulting from each study.

3. Results

3.1. Radiographic measurements of the forearm bones

The results of radiographic measurements are shown in Table 1.

3.2. Geometric study

3.2.1. Study 1

The boundary lines of Reg(ϕ) for $10° \leqslant \phi \leqslant 90°$ are shown in Fig. 6. The boundary lines consisted of two parts, the concave circular boundary derived from the equations expressing the impingement between circle *R* and circle *U* and the convex circular boundary derived from the equations expressing restriction by the string. The size of each Reg(ϕ) depended on the allowed angle of pronation of circle *R*. Reg($-\phi$) was a reflection of Reg(ϕ) on the *y*-axis. When pronation or supination of circle *R* was limited by impingement on circle *U*, rotation toward the opposite side was limited by the string and this limitation was far greater (Fig. 7). The boundary lines of Reg($-\phi/\phi$) were derived from equations that expressed the restriction by the string only. When circle *R* was allowed to pronate and supinate to 40° or more, each Reg($-\phi/\phi$) was in almost the same range which was 2 cm long in the radioulnar and 0.8 cm wide in the anteroposterior direction (Fig. 8).

3.2.2. Study 2

In study 2A, when *a* was reduced by 2.5, 5.0, and 7.5 mm, the size of each Reg(−70°/70°) was larger than the initial size of Reg(−70°/70°). When *a* was reduced by 10 mm, the size of Reg(−70°/70°) was smaller in the anteroposterior direction compared to the initial size of Reg(−70°/70°) because of the impingement between circle *R* and circle *U* (Fig. 9A). In study 2B, the size of each Reg(−70°/70°) was smaller than the initial size of Reg(−70°/70°). When *a* was reduced by 2.5 mm, the size of Reg(−70°/70°) in study 2B decreased by more than 30% (Fig. 9B).

3.2.3. Study 3

The boundary lines of Ang($-\phi/\phi$) for $10° \leqslant \phi \leq 90°$ are shown in Fig. 10. When the radius was allowed

Table 1
Radiographic measurements of the forearm bones

L	*B*	*S*	*T*	*M*	*N*
22.93 (0.93)	1.59 (0.38)	0.92 (0.42)	2.35 (0.24)	1.05 (0.33)	3.07 (0.14)

Data are given as mean values (SD) in centimeter.

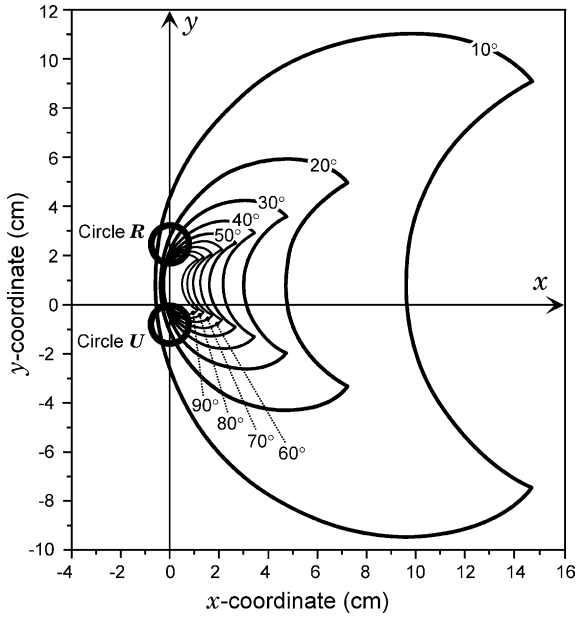


Fig. 6. Boundary lines of Reg(ϕ) for $10° \leqslant \phi \leqslant 90°$ at 10° intervals.

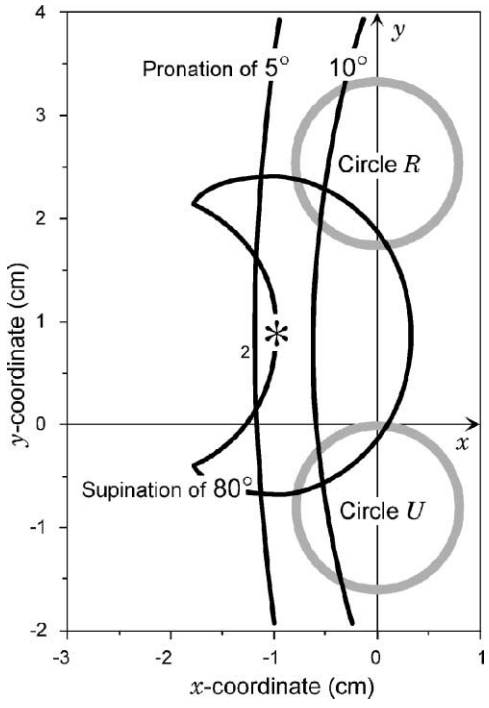


Fig. 7. Boundary lines of Reg(−80°), Reg(10°) and Reg(5°). When the AFR positioned at the asterisk, supination was limited to 80° by impingement between circle R and circle U and pronation was limited to less than 10° by the string.

to pronaste and supinate from its neutral rotation to 40° or more, each Ang($-\phi/\phi$) showed almost the same range, that is, about 14° radially, 7° ulnarly, 5° anteriorly and 4° posteriorly.

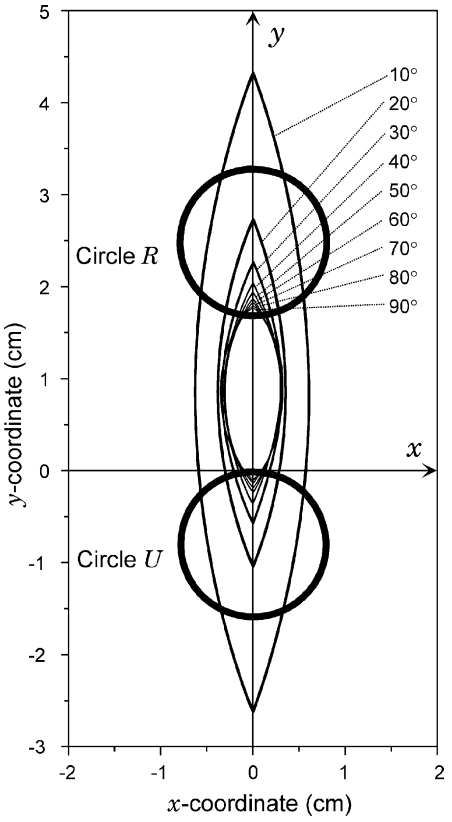


Fig. 8. Boundary lines of Reg($-\phi/\phi$)for $10° \leqslant \phi \leqslant 90°$ at 10° intervals.

4. Discussion

We obtained the region of the AFR around which the radius was allowed to rotate ϕ or more, and this region was represented by function Reg(ϕ). We thought the positional relationship between Reg(ϕ) and the AFR would indicate the range of forearm rotation. Deviation of the AFR from Reg(ϕ) would occur in the following three cases; the position of the AFR was changed without changing the size of Reg(ϕ); the AFR was deviated by decreasing the size of Reg(ϕ) and changing of the bones position; and, the AFR was deviated by decreasing the size of Reg(ϕ) without changing the bones position. The first case would be applied to the deformity of both bones of the forearm; the second case would be applied to narrowing of the interosseous space with shortening of the IOM; and the third case would be applied to a decrease of the slackness of the IOM. These three cases have been reported as the causes of limitation of forearm rotation associated with fracture of the forearm bones [1–4,8].

Tarr et al. [4] investigated the limitation of forearm rotation due to angular deformity of the forearm bones, and noted that soft-tissue tension, especially of the IOM, rather than impingement between the forearm bones, limited the rotation of the forearm in a cadaver study. Also in our study, when pronation or supin-

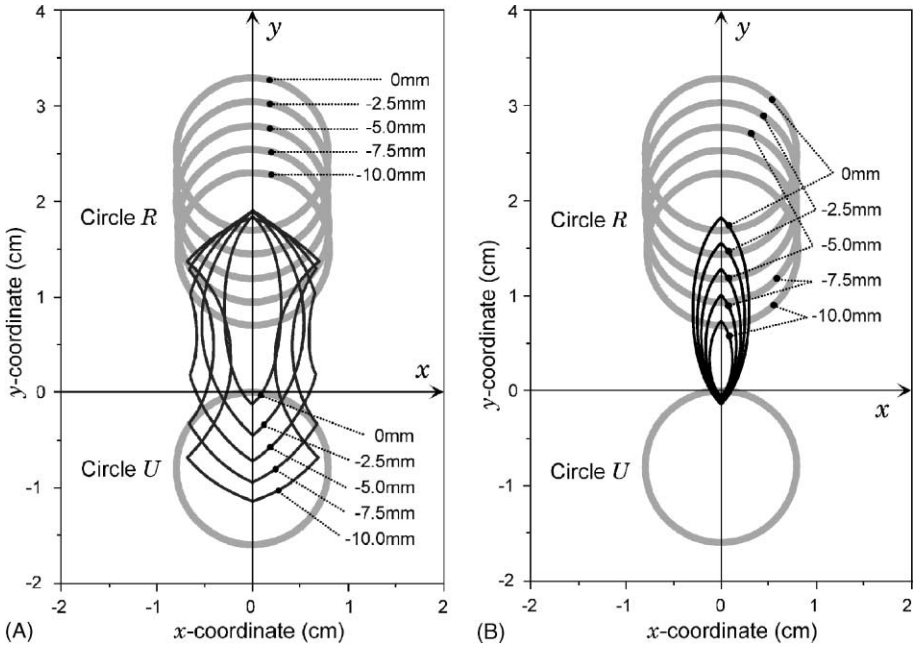


Fig. 9. Effects of the reduction of the distance between circle R and circle U on the size of Reg($-70°/70°$). (A) Results of study 2A. The length of the string was constant. (B) Results of study 2B. The length of the string was shortened in proportion to the reduction of the distance between circle R and circle U.

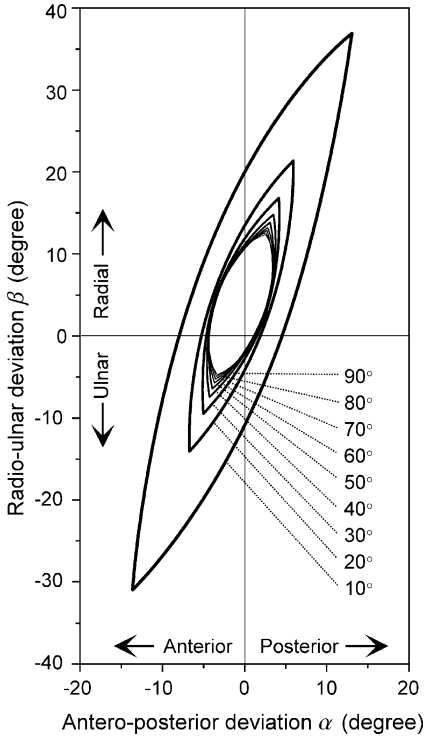


Fig. 10. Boundary lines of Ang($-\phi/\phi$) for $10° \leqslant \phi \leqslant 90°$ at $10°$ intervals.

ation of the forearm was limited by impingement of the forearm bones, rotation toward the opposite side was limited by the IOM and this limitation was far more significant. This result indicated that restriction by the IOM would be a major cause of limitation of the forearm rotation in our model with an angular deformity of the forearm bones.

Narrowing of the interosseous space between the radius and ulna has also been reported as a cause of limited forearm rotation [3]. In our study, narrowing of the interosseous space associated with shortening of the IOM always decreased the size of Reg($-\phi/\phi$) due to restriction by the IOM, however, narrowing of it without shortening of the IOM decreased the size of Reg($-\phi/\phi$) by impingement of forearm bones in particular when narrowing was more than 10 mm. Therefore, shortening of the IOM as a result of scarring along the IOM or a similar occurrence at the site of narrowing of the interosseous space was found to cause a decrease of forearm rotation. And narrowing of the interosseous space by more than 10 mm also would cause a decrease of forearm rotation regardless of the length of the IOM.

Price et al. [8] reported that decrease of forearm rotation might be due to tension of the IOM produced by soft tissue scarring. Scarring along the IOM would produce a reduction in the slackness of the IOM, which in turn would cause a decrease of forearm rotation. When the IOM was not loose in the forearm model with deformity in the anteroposterior direction, the forearm rotation was limited to just one side, either in pronation or supination.

As for the angle of deformity leading to significant loss of forearm rotation, this was 5° to the anterior side and 4° to the posterior side in our study. These angles were smaller than those reported in previous cadaver studies which stated that angulation of the forearm of

more than 15° or 20° might lead to significant loss of forearm rotation [1,4]. The possible reasons for the differences between the results of cadaver studies and ours are as follows. The slackness of the string which represented the IOM in our study might be lower than the actual one, because the tensile strain of the central band of the IOM, the strongest structure in the IOM, was applied to determine the slackness of the string; the laxity of the adjacent joints was not taken into account in the model of our study, while actual forearm rotation produces not only a rotational movement but also a sliding movement of the distal radioulnar joint [6,9,10]. If the sliding movement of the distal radioulnar joint had been included in the model used in this study, the position of the AFR would translate from $\mathrm{Reg}(-\phi/\phi)$ to $\mathrm{Reg}(-\phi - \Delta\phi/\phi + \Delta\phi)$ $(\Delta\phi \geqslant 0)$ where the forearm was allowed to rotate more widely. This phenomenon means the size of each $\mathrm{Ang}(-\phi/\phi)$ would increase in the anteroposterior direction.

Translation of the AFR at the maximum rotation is also important in the patients with deformity of the forearm bones. When further rotational load is exerted on the radius, which is restricted by the IOM, translation of the AFR or tear of the IOM was required for rotation of the radius. And, when the translation of the axis reached the normal limit, an eccentric load would be exerted on the adjacent joints or the IOM at maximum pronation or supination. We have previously reported a case of anterior dislocation of the radial head associated with limitation of forearm rotation caused by a malunited fracture of the forearm bones [5], and we think the eccentric load which led to the dislocation of it was produced by the above mechanism.

5. Conclusion

The positional relationship between the AFR, forearm bones with the IOM played an important role in forearm rotation. In our models, the main cause of limitation of the forearm rotation in the case of an angular deformity of the forearm bones was restriction by the IOM. Limitation of forearm rotation by narrowing of the interosseous space was affected by shortening of the IOM which simulated scarring along the IOM. In the 3D model without sliding movement of the adjacent joints, the direction and magnitude of the angular deformity which would lead to limitation of forearm rotation were determined.

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